Can we find factorial of irrational and rational numbers also only for intergers or other than that

Very good question 👍

Most students think factorial is only for integers, but actually it can be extended beyond integers too.

Let's understand step by step.

1️⃣ Factorial for Integers (Basic Definition)

For a non-negative integer:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 1$$

Examples:

- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- $3! = 6$
- $0! = 1$

👉 This definition works only for non-negative integers.

2️⃣ Can We Find Factorial of Rational Numbers?

Example:

$$\left(\frac{1}{2}\right)! \quad ?$$

Using normal multiplication definition ❌ Not possible.

But using a special function called the:

⭐ Gamma Function

Gamma function

The Gamma function extends factorial to non-integer values.

Important Formula:

$$n! = \Gamma(n+1)$$

So for non-integers:

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right)$$

And:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

So:

$$\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$$

✅ So factorial can be found for rational numbers using Gamma function.

3️⃣ Can We Find Factorial of Irrational Numbers?

Example:

$$\sqrt{2}! \quad ?$$

Using Gamma function:

$$\sqrt{2}! = \Gamma(\sqrt{2}+1)$$

✅ Yes — factorial is defined for irrational numbers too
(as long as it's not a negative integer).

4️⃣ Where Factorial is NOT Defined

Gamma function is not defined for:

$$-1, -2, -3, -4, \ldots$$

So factorial does NOT exist for negative integers.

Example:

$$(-3)! \quad ❌ \text{ Not defined}$$

5️⃣ Summary Table

Type of Number	Factorial Exists?	How?
Positive Integer	✅ Yes	Normal definition
Zero	✅ Yes	0! = 1
Rational (non-integer)	✅ Yes	Gamma function
Irrational	✅ Yes	Gamma function
Negative Integer	❌ No	Not defined

6️⃣ Why Students Think It's Only for Integers?

Because in school mathematics, factorial is defined only as repeated multiplication — which only makes sense for integers.

But higher mathematics extends it using calculus.

If you want, I can also:

- Show graph of Gamma function 📈
- Explain why $\Gamma(1/2) = \sqrt{\pi}$
- Explain factorial of negative fractions

Just tell me 😊

What is the name of symbol we use un gamma function also explain step by step with examples

Good question 👍

1️⃣ Name of the Symbol Used in Gamma Function

The symbol used is:

$$\Gamma$$

It is called capital Gamma.

It comes from the Greek alphabet.

- Small gamma: γ
- Capital gamma: Γ

The Gamma function is written as:

$$\Gamma(x)$$

2️⃣ What is the Gamma Function?

The Gamma function is a function that extends factorial to non-integer numbers.

Important Relation:

$$n! = \Gamma(n+1)$$

So factorial is just a special case of Gamma function.

3️⃣ Step-by-Step Understanding

Step 1: Definition of Gamma Function

The Gamma function is defined as:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt$$

This is an improper integral (goes to infinity).

Don't worry — we won't solve the full integral manually, just understand how it works.

4️⃣ Example 1: Prove Γ(1) = 1

Put $x = 1$:

$$\Gamma(1) = \int_0^\infty t^{1-1} e^{-t} dt$$

$$= \int_0^\infty e^{-t} dt$$

Now integrate:

$$\int e^{-t} dt = -e^{-t}$$

Apply limits:

$$[-e^{-t}]_0^\infty$$

At $t = \infty$: $e^{-\infty} = 0$
At $t = 0$: $e^0 = 1$

So:

$$0 - (-1) = 1$$

✅ Therefore:

$$\Gamma(1) = 1$$

And since:

$$0! = 1$$

It matches.

5️⃣ Example 2: Prove $\Gamma(2) = 1!$

$$\Gamma(2) = \int_0^\infty t^{2-1} e^{-t} dt$$

$$= \int_0^\infty t e^{-t} dt$$

Using integration by parts (result):

$$\Gamma(2) = 1$$

And:

$$1! = 1$$

Correct again ✅

6️⃣ Example 3: Half Factorial

Now interesting case:

$$\Gamma\left(\frac{1}{2}\right)$$

This integral gives:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

So:

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right)$$

And:

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

So:

$$\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$$

Very surprising result 😄

7️⃣ Important Property (Very Important)

Gamma function satisfies:

$$\Gamma(x+1) = x\Gamma(x)$$

This behaves exactly like factorial:

$$(n+1)! = (n+1)n!$$

That is why Gamma extends factorial.

8️⃣ Where Gamma is NOT Defined

It is undefined for:

$$-1, -2, -3, \ldots$$

Because the integral blows up (becomes infinite).

9️⃣ Simple Summary

- Symbol: Γ (capital Gamma)
- It is a Greek letter
- Used to extend factorial
- Works for:
  - Integers
  - Fractions
  - Irrational numbers
- Not defined for negative integers

If you want, I can next explain:

- Why $\Gamma(1/2) = \sqrt{\pi}$ (beautiful trick using polar coordinates)

- Graph of Gamma function
- Gamma for negative fractions

Just tell me 😊

What is he integration of e power negative x also which formula using

Good 👍 let's do it clearly step by step.

1️⃣ Integration of e^{-x}

We want:

$$\int e^{-x}\,dx$$

🔹 Formula Used

We use the basic exponential integration rule:

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$$

(where a is a constant)

2️⃣ Apply the Formula

Here:

$$e^{-x} = e^{(-1)x}$$

So $a = -1$

Using the formula:

$$\int e^{-x}dx = \frac{1}{-1}e^{-x} + C$$

$$= -e^{-x} + C$$

✅ Final Answer:

$$\boxed{\int e^{-x}dx = -e^{-x} + C}$$

3️⃣ Why This Works (Derivative Check)

Differentiate your answer:

$$\frac{d}{dx}(-e^{-x})$$

Using chain rule:

Derivative of e^{-x} is:

$$-e^{-x}$$

So:

$$\frac{d}{dx}(-e^{-x}) = -(-e^{-x}) = e^{-x}$$

✔ Correct!

4️⃣ Alternative Method (Substitution Method)

Let:

$$u = -x$$

Then:

$$du = -dx$$

So:

$$\int e^{-x} dx$$

becomes

$$\int e^{u}(-du)$$

$$= -\int e^{u} du$$

$$= -e^{u} + C$$

Substitute back $u = -x$:

$$= -e^{-x} + C$$

Same result ✅

If you want, I can also explain:

- Integration of e^{-ax}
- Integration of xe^{-x} (used in Gamma function)
- Why e^{x} is special in calculus

Just tell me 🙂

If there is any function in e power

Good question 👍

You're asking:

👉 What if there is a function in the power of e?

Like:

$$\int e^{f(x)} dx$$

Let's understand step by step.

1️⃣ Basic Rule (Very Important)

If the exponent is a function:

$$\int e^{f(x)} dx$$

Then we use substitution.

But there is one important condition:

> The derivative of the inside function must also be present.

2️⃣ Main Formula

If:

$$\frac{d}{dx}(f(x)) = f'(x)$$

Then:

$$\int e^{f(x)} f'(x) dx = e^{f(x)} + C$$

This comes from the chain rule.

3️⃣ Example 1 (Easy Case)

$$\int e^{3x} dx$$

Here:

$f(x) = 3x$

Derivative:

$$f'(x) = 3$$

So we divide by 3:

$$= \frac{1}{3}e^{3x} + C$$

4️⃣ Example 2 (Substitution)

$$\int e^{x^2}(2x)dx$$

Let:

$$u = x^2$$

Then:

$$du = 2xdx$$

Now integral becomes:

$$\int e^u du$$

$$= e^u + C$$

Substitute back:

$$= e^{x^2} + C$$

5️⃣ Important Case ⚠

Now look at:

$$\int e^{x^2} dx$$

Here derivative of x^2 is $2x$

But we don't have $2x$.

❌ This integral cannot be solved using elementary functions.

It is expressed using a special function.

6️⃣ General Rule Summary

Case I:

$$\int e^{ax}dx = \frac{1}{a}e^{ax} + C$$

Case 2:

$$\int e^{f(x)}f'(x)dx = e^{f(x)} + C$$

Case 3:

If derivative is NOT present → use substitution or special functions.

7️⃣ Why e^x is Special?

Because:

$$\frac{d}{dx}(e^x) = e^x$$

It remains the same after differentiation.

That's why exponential functions are very powerful in calculus.

If you want, I can next explain:

- How to integrate xe^{-x} (used in Gamma function)
- Why $\int e^{x^2}dx$ has no simple answer
- How chain rule connects differentiation and integration

Just tell me 😊

Ok still confusion about gamma function if we simply find the factorial of sqrt or fraction functions which steps we follow in easy and simple way

Very good 👍 this confusion is normal.
Let's clear it in the most simple way possible.

You are asking:

> If I want factorial of √2 or 1/2 or any fraction, what steps do I follow?

⭐ Step 1: Use the Main Formula

Factorial for non-integers is defined using the
Gamma function

The rule is:

$$x! = \Gamma(x+1)$$

That's it. This is the starting step always.

⭐ Step 2: Add 1 to the Number

If you want:

Example 1: $\left(\frac{1}{2}\right)!$

Add 1:

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right)$$

Example 2: $\sqrt{2}!$

Add 1:

$$\sqrt{2}! = \Gamma(\sqrt{2}+1)$$

⭐ Step 3: Now What?

There are 3 possibilities:

✅ Case A: It reduces to a known Gamma value

Some special Gamma values are known:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Very important identity:

$$\Gamma(x+1) = x\Gamma(x)$$

This behaves like factorial rule.

🔹 Example: Find $\left(\frac{1}{2}\right)!$

We got:

$$\Gamma\left(\frac{3}{2}\right)$$

Use identity:

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

We know:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

So:

$$\left(\frac{1}{2}\right)! = \frac{1}{2}\sqrt{\pi}$$

Done ✅

No integration needed.

⭐ Step 4: If It Is Something Like √2!

Example:

$$\sqrt{2}! = \Gamma(\sqrt{2}+1)$$

This cannot be simplified nicely.

So what do we do?

👉 We calculate using:

- Scientific calculator
- Computer
- Approximation tables

There is no simple closed formula like √π.

🔴 Important Truth

In practice:

✔ We rarely calculate Gamma integral manually

✔ We use properties

✔ Or use calculator/computer

Because the Gamma definition:

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$$

is very difficult to solve directly.

⭐ So The SIMPLE METHOD Is:

If integer:

Use normal factorial.

If fraction like 1/2:

1. Add 1
2. Use Gamma identity
3. Use known value $\Gamma(1/2)=\sqrt{\pi}$

If irrational like $\sqrt{2}$:

1. Add 1
2. Use calculator to compute $\Gamma(x+1)$

🧠 Why School Doesn't Teach This?

Because:

- Requires calculus
- Requires improper integrals
- Requires advanced mathematics

It is usually taught in higher mathematics.

📌 Final Summary (Very Simple)

To find factorial of non-integer:

$$\boxed{x! = \Gamma(x+1)}$$

Then:

- Use Gamma properties
- Or numerical approximation

If you want, I can now explain:

- Why Γ(1/2) = √π (very beautiful proof)
- Or show how calculator computes Gamma
- Or graph of Gamma function

Tell me which one 😊

Ok let me know how we knew which function are equal to sqrt pie and also how we calculate the factorial of sqrt functions

Excellent question 👏 now we are going deeper.

You are asking two things:

1️⃣ How do we know which Gamma value equals √π?
2️⃣ How do we calculate factorial of √ functions like √2! ?

Let's go slowly and clearly.

🔵 Part 1: Why is Γ(1/2) = √π ?

We don't guess it.

It comes from the definition of the
Gamma function

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$$

Step 1: Put x = 1/2

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} e^{-t} dt$$

That means:

$$\int_0^\infty \frac{e^{-t}}{\sqrt{t}} dt$$

Step 2: Clever Substitution

Let:

$$t = y^2$$

Then:

$$dt = 2y dy$$

Now substitute:

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{e^{-y^2}}{y} (2y dy)$$

The y cancels:

$$= 2\int_0^\infty e^{-y^2} dy$$

Step 3: Very Famous Integral

The integral

$$\int_0^\infty e^{-y^2} dy$$

is a very famous result in mathematics.

It equals:

$$\frac{\sqrt{\pi}}{2}$$

So:

$$\Gamma\left(\frac{1}{2}\right) = 2 \times \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$$

✔ That is how we KNOW it equals √π.

It is mathematically proven using double integrals and polar coordinates.

🔵 Part 2: How Do We Calculate √2! ?

Now suppose we want:

$$\sqrt{2}!$$

Step 1: Use Main Rule

$$x! = \Gamma(x+1)$$

So:

$$\sqrt{2}! = \Gamma(\sqrt{2}+1)$$

Step 2: Can We Simplify?

We check:

Does √2 + 1 reduce to 1/2 ?

No.

Is it integer?

No.

So no simple formula like √π.

Step 3: What Do We Do?

We compute numerically.

Using calculator or software:

$$\Gamma(2.4142...)$$

Approximate value:

$$\sqrt{2}! \approx 1.354$$

There is no simple closed form.

🔵 Important Understanding

Only special values like:

- Γ(1/2)
- Γ(3/2)
- Γ(5/2)

reduce nicely because we can repeatedly use:

$$\Gamma(x+1) = x\Gamma(x)$$

Example:

$$\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \frac{1}{2}\sqrt{\pi}$$

🔵 When Does √π Appear?

Whenever we reduce Gamma down to:

$$\Gamma(1/2)$$

That's the only place √π comes from.

🧠 Final Clear Picture

• √π comes from solving a special Gaussian integral

• Only half-integers give nice √π answers

• Irrational numbers like √2 give messy decimal values

• We calculate them numerically

If you want, I can next explain:

- Why the Gaussian integral equals √π
- Or show graph of Gamma function
- Or explain half-integer factorial pattern clearly

Tell me 😊

Means if numerator is odd and denominator is even or 2 then it is equal to sqrt pie also how we compute sqrt factorial numerically means hwat is the process if i want to write a program for it firstbi have to know what is the mathematical process also what is the proof.of integraion of e power negative y square is equal to sqrt pie by 2

Excellent 👏 now you are asking the right deep questions.

We will answer clearly in 3 parts:

1️⃣ When do we get √π ?

2️⃣ How to compute factorial of √2 numerically (mathematical process)

3️⃣ Why ∫ e^(-y²) dy = √π / 2

🔵 PART I

When Does √π Appear?

You asked:

If numerator is odd and denominator is 2, does √π always appear?

Yes ✅ exactly.

Whenever the Gamma function argument is a half-integer:

$$\Gamma\left(\frac{1}{2}\right), \Gamma\left(\frac{3}{2}\right), \Gamma\left(\frac{5}{2}\right), \Gamma\left(\frac{7}{2}\right)$$

That means:

$$\frac{\text{odd number}}{2}$$

Then the result always contains √π.

Why?

Because we reduce everything to:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Using property:

$$\Gamma(x+1) = x\Gamma(x)$$

Example:

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\sqrt{\pi}$$

So yes — odd/2 always leads to √π.

🔵 PART 2

How Do We Compute √2! Numerically?

We know:

$$\sqrt{2}! = \Gamma(\sqrt{2}+1)$$

Now mathematically Gamma is defined as:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$$

So for √2:

$$\Gamma(\sqrt{2}+1) = \int_0^\infty t^{\sqrt{2}} e^{-t} dt$$

💡 Numerical Process

Since this integral has no simple solution, we approximate it.

Main numerical methods:

1️⃣ Trapezoidal Rule
2️⃣ Simpson's Rule
3️⃣ Gaussian Quadrature
4️⃣ Lanczos Approximation (used in programming libraries)

🔵 Most Common Method in Programming

Most programming languages use:

⭐ Lanczos Approximation

It approximates Gamma using:

$$\Gamma(z) \approx \sqrt{2\pi}(z+g-1/2)^{z-1/2} e^{-(z+g-1/2)} A(z)$$

(where A(z) is a coefficient polynomial)

You don't compute the integral directly —
you use approximation formula.

That's how calculators compute √2!.

🔵 PART 3

Why Does

$$\int_0^\infty e^{-y^2} dy = \frac{\sqrt{\pi}}{2}$$

This is a famous proof.

Step I: Let

$$I = \int_0^\infty e^{-y^2} dy$$

We cannot integrate directly.

So we square it.

Step 2: Square the integral

$$I^2 = \left(\int_0^\infty e^{-x^2} dx \right) \left(\int_0^\infty e^{-y^2} dy \right)$$

Combine:

$$I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$$

Now this is a double integral.

Step 3: Convert to Polar Coordinates

We use:

$$x^2 + y^2 = r^2$$

Area element becomes:

$$dx dy = r dr d\theta$$

Limits:

$$r : 0 \to \infty$$

$$\theta : 0 \to \frac{\pi}{2}$$

Because we are in first quadrant.

Step 4: Rewrite Integral

$$I^2 = \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r dr d\theta$$

Step 5: Solve Inner Integral

Let:

$$u = r^2$$

Then:

$$du = 2r dr$$

So:

$$\int_0^{\infty} e^{-r^2} r dr = \frac{1}{2}$$

Step 6: Solve Outer Integral

$$I^2 = \int_0^{\pi/2} \frac{1}{2} d\theta = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

So:

$$I = \frac{\sqrt{\pi}}{2}$$

✔ Proven.

🔵 Final Understanding

✔ Half-integers → always contain √π

✔ Irrational numbers → use numerical approximation

✔ √π comes from Gaussian double integral proof

✔ Programming uses Lanczos approximation

If you want, I can next:

- Show you how to implement Gamma in C++
- Or derive full Gaussian integral proof more slowly
- Or explain why negative integers are undefined

Tell me which one 😊

Also ehat if we don't have numerator odd and denominator 2 or even then

Very good 👏 this is the final missing piece.

You're asking:

> What if the number is NOT of the form (odd / 2)?
> Then what happens? Do we still get √π?

Short answer:

❌ No, we do NOT get √π in a simple form.

✔ We use Gamma normally, but it won't simplify nicely.

Now let's understand properly.

🔵 Case 1: Half-Integers (odd/2)

Example:

$$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}$$

These always reduce to:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

So they contain √π.

🔵 Case 2: Other Fractions (Not odd/2)

Example:

$$\frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}$$

These do NOT reduce to √π.

Example:

$$\Gamma\left(\frac{1}{3}\right)$$

This has no simple form using π or √π.

It is just a special constant (decimal value).

So:

$$\left(\frac{1}{3}\right)! = \Gamma\left(\frac{4}{3}\right)$$

This is just a decimal number.

🔵 Case 3: Irrational Numbers

Example:

$$\sqrt{2}!, \pi!, e!$$

We use:

$$x! = \Gamma(x+1)$$

But these do NOT simplify.

We compute numerically.

🔵 Why Only Half-Integers Give √π?

Because √π comes from the special integral:

$$\int_0^\infty e^{-y^2} dy$$

That integral appears ONLY when we evaluate:

$$\Gamma\left(\frac{1}{2}\right)$$

Other fractions produce integrals like:

$$\int_0^\infty t^{-2/3} e^{-t} dt$$

which do not simplify to π.

🔵 Big Picture

Type of number	Result form
Integer	Normal factorial
Odd / 2	Contains √π
Other fractions	Special decimal value
Irrational	Numerical value

🔵 Important Understanding

Gamma function works for ALL positive real numbers.

But:

Only special inputs give "beautiful" answers.

Others give complicated constants.

🔵 Simple Example Comparison

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

But:

$$\Gamma\left(\frac{1}{3}\right)$$

is just:

≈ 2.6789...

No π appears.

🧠 Final Conclusion

If numerator is NOT odd and denominator NOT 2:

✔ Factorial still exists

❌ But no √π

✔ Value is computed numerically

✔ It's still perfectly valid

If you want, I can now explain something very interesting:

There is a formula that connects Gamma and π even for other fractions — it's called Euler's reflection formula.

It's beautiful.

Tell me if you want to see it 😊